Image Generation With — WGANs, BiGANs, and Domain Adversarial Networks

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October 16, 2025

1 Introduction

This self assignment builds on your GAN work and explores three influential adversarial paradigms on the **same dataset/splits** used previously: **Wasserstein GANs (WGAN/WGAN-GP)**, **Bi-directional GANs (BiGAN)**, and **Domain Adversarial Neural Networks (DANN)** for domain adaptation. Follow the same *reporting style, preprocessing, and logging conventions* as your earlier files.

2 Dataset & Preprocessing

Use identical train/val/test splits and normalization statistics as before. For **DANN**, designate *source* and *target* domains (e.g., different augmentations or sub-domains of your dataset) and maintain a consistent evaluation protocol across models.

3 Task Objectives

1. Implement and train **WGAN** (with weight clipping) and **WGAN-GP** (with gradient penalty) using both MLP and DCGAN-style backbones.
2. Implement and train a **BiGAN** (encoder + generator + discriminator) to learn *inference* (mapping $x \to z$) jointly with generation ($z \to x$).
3. Implement and train a **DANN** for unsupervised domain adaptation: learn domain-invariant features using a gradient-reversal layer (GRL) while optimizing source label prediction.
4. Evaluate with appropriate metrics (FID/KID/IS for generative models; target-domain accuracy for DANN) and provide qualitative visualizations.

4 Model Specifications

4.1 WGAN / WGAN-GP

Critic f_θ outputs a real-valued score (no sigmoid); **Generator** G maps $z \sim \mathcal{N}(0, I)$ to images.

- **WGAN loss** (weight clipping c): maximize $\mathbb{E}[f(x)] - \mathbb{E}[f(G(z))]$ for critic; minimize $-\mathbb{E}[f(G(z))]$ for generator.

- **WGAN-GP loss**: add gradient penalty $\lambda\ \mathbb{E}_{\hat{x}}\big(\|\nabla_{\hat{x}} f(\hat{x})\|_2 - 1\big)^2$ with $\hat{x}$ sampled on lines between real and generated samples.
- **Updates**: typically $n_{\text{critic}} \in \{1, 5\}$ critic steps per generator step; **Adam** with $\alpha = 1\text{e}{-4}$, $\beta_1 = 0.0$, $\beta_2 = 0.9$ (recommended).

4.2 BiGAN

Learn **encoder** $E : x \to z$, **generator** $G : z \to x$, and **discriminator** D on joint pairs: real $(x, E(x))$ vs fake $(G(z), z)$.

- **Objective**: standard GAN objective on joint space; D predicts whether a pair is real or fake. Optimize E and G together against D.
- **Backbones**: DCGAN-style G and D; E mirrors D but maps to z (mean-only or mean+logvar if trying a variational extension).
- **Evaluation**: image reconstruction via $G(E(x))$, latent interpolations, downstream classification using $E(x)$ features (optional).

4.3 DANN (Domain Adversarial Network)

Model comprises: **feature extractor** F, **label predictor** C (trained with source labels), and **domain classifier** D coupled through a **gradient reversal layer (GRL)**.

- **Loss**: $\mathcal{L} = \mathcal{L}_{\text{cls}}(C(F(x_s)), y_s) - \lambda\ \mathcal{L}_{\text{dom}}(D(F(x)), d)$ where $d \in \{0, 1\}$ indicates source/target; GRL multiplies the gradient by $-\lambda$ for F.
- **Protocol**: unlabeled target data participates only in $\mathcal{L}_{\text{dom}}$. Report source/train accuracy and **target-domain test accuracy**.
- **Schedulers**: ramp λ from 0 to a max value over epochs; optionally use *Entropy Minimization* or *Confidence Thresholding* on target (optional).

5 Self Evaluation Protocol

5.1 Generative (WGAN/WGAN-GP/BiGAN)

- **Quantitative**: FID, KID, IS (if applicable). If unavailable, report feature-space NN distances or precision/recall for generative models.
- **Qualitative**: fixed-noise grids over epochs, latent interpolations; BiGAN reconstructions $G(E(x))$.

5.2 Domain Adaptation (DANN)

- **Target accuracy**: main metric on the target test set.
- **Diagnostics**: source vs target feature visualizations (e.g., t-SNE), confusion matrices, per-class target accuracy.

6 References

- M. Arjovsky, S. Chintala, L. Bottou, "Wasserstein GAN," ICML 2017.
- I. Gulrajani et al., "Improved Training of Wasserstein GANs," NeurIPS 2017.
- J. Donahue, P. Krähenbühl, T. Darrell, "Adversarial Feature Learning" (BiGAN), ICLR 2017.
- Y. Ganin, V. Lempitsky, "Unsupervised Domain Adaptation by Backpropagation" (DANN), ICML 2015.